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PICTURED BELOW IS WHAT THE AVERAGE STAT BLOCK LOOKS LIKE IN THE DATASET

DnD 5e Challenge Rating Predictor

Large construct (Jupyter Notebook), True Neutral

Armor Class 1
Hit Points 1 (1d10 - 5)
Speed Immobile. ~14 min runtime

STR	DEX	CON	INT	WIS	CHA
1 (-5)	1 (-5)	1 (-5)	40 (+15)	40 (+15)	40 (+15)

Saving Throws Int +17, Wis +17
Skills Insight +17, Investigation +17, Perception +17
Damage Vulnerabilities high cr monsters; textual analysis
Damage Immunities bludgeoning, piercing, and slashing from nonmagical attacks; low cr monsters
Condition Immunities incapacitated
Senses passive Perception 27
Languages English, telepathy 5 ft.
Challenge 2 (Predicted by itself)

Legendary Resistance (3/Day). If the notebook fails a saving throw, it can choose to succeed instead.

Legendary Actions

The notebook can take 3 legendary actions, choosing from the options below. Only one legendary action option can be used at a time and only at the end of another creature's turn. The notebook regains spent legendary actions at the start of its turn.

Running Cells (Costs 4 Actions). : The Notebook runs all cells in order, charming creatures within 5 feet of it. The target must succeed on a DC 18 Wisdom saving throw against this magic or become charmed for 1 minute. The charmed target can repeat the saving throw at the end of each of its turns, ending the effect on itself on a success. If a target's saving throw is successful or the effect ends for it, the target is immune to the notebook's effects for the next 24 hours.

GOAL

In Dungeons & Dragons 5th Edition, Challenge Rating (CR) is a numerical estimate of a monster's combat difficulty. It determines whether an encounter is trivially easy or potentially lethal for a given party level.

Calculating it correctly involves using a multi-step formula from the Dungeon Master's Guide. This is a process that is **tedious and error-prone** for custom homebrew content.

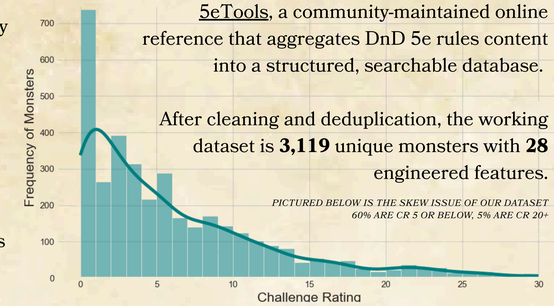
This project trains supervised machine learning models to predict CR directly from a monster's raw statistics, giving Dungeon Masters an instant balance check tool .

DATASET

The raw dataset contains 4,454 monsters from 5eTools, a community-maintained online reference that aggregates DnD 5e rules content into a structured, searchable database.

After cleaning and deduplication, the working dataset is **3,119** unique monsters with **28** engineered features.

PICTURED BELOW IS THE SKEW ISSUE OF OUR DATASET
60% ARE CR 5 OR BELOW, 5% ARE CR 20+



METHODOLOGY

Raw Data
4,454 MONSTERS
↓
Cleaning & Deduplication
SOURCE PRIORITY RANKING, HP/AC EXTRACTION
↓
EDA
CORRELATION ANALYSIS, DISTRIBUTION CHECKS
↓
Feature Engineering
28 FEATURES: SIZE_RANK, TYPE ONE-HOT, IS_LEGENDARY, LOG_HP, ETC.
↓
Preprocessing
LOG TRANSFORM, Z-SCORE, MIN-MAX SCALING
↓
Stratified Split
80/20 CLASSICAL
70/15/15 NEURAL NETWORK
↓
Model Training
RANDOM FOREST, XGBOOST, MLP NEURAL NETWORK
↓
Evaluation
BEST MODEL: XGBOOST
SUMMARY, CONCLUSION, FUTURE WORKS

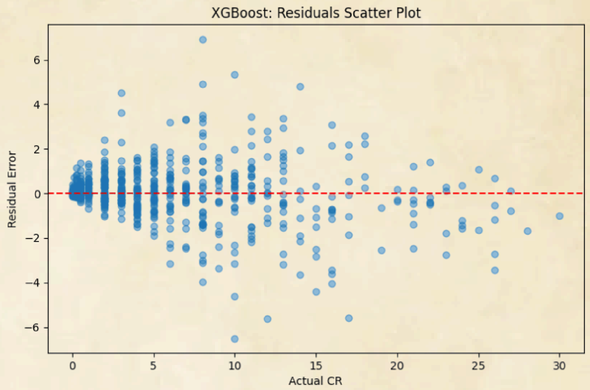
RESULTS

- XGBoost predicts within 1 CR ~71% of the time
- HP (r = 0.93) is the dominant predictor – survivability drives CR
- All models degrade at CR 15+: error nearly doubling vs low-tier monsters
- High-CR monsters rely on offensive mechanics (damage output, multiattack) not captured in raw stats alone

MODEL	MAE	RMSE	R^2	WITHIN 1 CR
Random Forest	0.885	1.355	95.1	68.8%
XGBoost	**0.85**	**1.3199**	**95.3**	**70.7%**
MLP Neural Net	0.907	1.383	94.9	70.3%

EVALUATED ON HELD-OUT TEST SET (N=468).
WITHIN 1 CR = % OF PREDICTIONS WITHIN ±1 OF ACTUAL CR.

CR Tier	RF MAE	XGB MAE	NN MAE
0-1 (Minions)	0.26	0.25	0.26
1-5 (Low)	0.69	0.65	0.68
5-10 (Mid)	1.31	1.36	1.34
10-20 (High)	1.68	1.56	1.62
20+ (Bosses)	1.55	1.28	1.80



FINDINGS

1. **XGBoost is recommended for homebrew CR validation,** achieving 70.7% accuracy within ±1 CR. A DM submitting a custom monster stat block gets a reliable CR estimate instantly.

2. **Survivability dominates CR prediction.** HP and Constitution alone drive most of the signal. Future work should incorporate average damage per round and attack bonus to close the gap at high tiers.

3. **Data imbalance is the primary bottleneck.** CR 20+ monsters represent only 5.3% of the dataset. Targeted data collection for high-tier monsters would likely produce the largest accuracy gains.

4. **Classical models outperform the neural network on structured tabular data at this scale.** With ~2,000 training samples, XGBoost's inductive bias toward decision boundaries is better suited than an MLP's general function approximation.

CONCLUSION

XGBoost is reliable enough for real homebrew use at low-to-mid CR. The main remaining gap is high-tier monsters, where damage output matters more than raw stats – future work should incorporate offensive mechanics to close this.

XGBOOST ARCHITECTURE

Trees: 250 estimators, Max depth: 5, learning rate: 0.1, Subsample: 0.8 (row subsampling per tree), Regularization: L1+L2 (built-in), Tuning: 5-Fold GridSearchCV (16 configs tested), Input: 27 engineered features.

Tuning improved MAE by 0.023 over the baseline